



University of Colombo, Sri Lanka



University of Colombo School of Computing



**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2025 — 1st Year Examination — Semester I

IT1106 — Information Systems

Multiple Choice Question Paper
(2 Hours)

Important Instructions

- The duration of the paper is 2 Hours.
- The medium of instructions and questions is English.
- This paper has 40 questions on 10 pages. Answer all questions.
- All questions are of the MCQ (Multiple Choice Questions) type.
- Each question will have 5 (five) choices with ONLY ONE correct answer.
- This paper consists of 100 marks and all the questions will carry equal marks.
- Answers should be marked on the special answer sheet provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/investigator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are not allowed.
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- 1) Which component of an Information System is responsible for transforming input data into useful information?

(a) Input	(b) Output	(c) Storage
(d) Processing	(e) Registering	

- 2) Which type of software acts as an interface between the user and computer hardware?

(a) Middleware	(b) System software	(c) Utility software
(d) Programming software	(e) Application software	

- 3) Consider the following two lists of “devices” and their “purposes” that have not been correctly matched.

Device	Purpose
A. Joystick	I. Capturing voice or sound.
B. Scanner	II. Reading coded product information.
C. Microphone	III. Entering printed text or images into the computer.
D. Barcode Reader	IV. Controlling movement in games or simulations.

Which of the following **correctly** matches the above listed devices with the purposes?

(a) A–II, B–III, C–IV, D–I	(b) A–III, B–II, C–I, D–IV	(c) A–IV, B–III, C–I, D–II
(d) A–I, B–III, C–II, D–IV	(e) A–II, B–IV, C–III, D–I	

- 4) According to Porter, the value chain refers to:

(a) a set of interrelated activities through which a firm creates value for its customers.
(b) a collection of support services provided by information systems to enhance business processes.
(c) the process of distributing products from manufacturers to retailers and retailers to customers.
(d) a network of suppliers, distributors, and customers connected through the internet to provide fast service.
(e) a sequence of computer-based information systems integrated to increase the performance of an organization.

- 5) Consider the following statement.

In cloud computing, the _____ licensing model allows organizations to pay only for the computing resources they actually use, without purchasing permanent software licenses.

Select the **most** appropriate term to fill in the blank in the above statement.

(a) Site	(b) Perpetual	(c) Pay-as-you-go
(d) Concurrent user	(e) Subscription-based	

6) Which of the following is **NOT** typically considered a core component of an information system?

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|--------------|--------------|--------------|
| (a) Hardware | (b) Security | (c) Services |
| (d) Data | (e) Networks | |

7) The major role of an information system in a business firm is to:

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| (a) improve operational efficiency by hardware maintenance and software updates. |
| (b) support customers to identify, select, and purchase their required products quickly. |
| (c) provide effective support to its business strategies for gaining competitive advantage. |
| (d) identify potential suppliers who can deliver raw materials on time in required quantities. |
| (e) collect, store, and share data and information as required by its employees and customers. |

8) Consider the following paragraph with three blanks.

The strategic role of information systems involves using _____ to develop products, services, and capabilities that give a company major advantages over _____ in the global marketplace. This role is achieved through a _____, which is a collection of information systems that supports or shapes a business's competitive position and strategies.

Which of the following completes the blanks in the above paragraph in the **correct** order?

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| (a) Daily operations; employees; management hierarchy |
| (b) Enterprise servers; suppliers; production workflow |
| (c) Networking hardware; market trends; operational database |
| (d) Software applications; customer feedback; user interface design |
| (e) Information technology; competitive forces; strategic information architecture |

9) A large retail organization is undergoing a digital transformation. Its executive team includes a Chief Executive Officer (CEO), a Chief Information Officer (CIO), and a Chief Technology Officer (CTO).

One executive, M, is responsible for ensuring that the organization's internal information systems are aligned with business objectives, enhance operational efficiency, and support managerial decision-making.

Another executive, N, is tasked with driving innovation by identifying and adopting emerging technologies, leading the development of new digital products, and strengthening customer-facing technological capabilities.

Which option **best** identifies the roles of M and N?

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|-------------------------------------|
| (a) M is the CTO, and N is the CEO. |
| (b) M is the CEO, and N is the CTO. |
| (c) M is the CIO, and N is the CTO. |
| (d) M is the CTO, and N is the CIO. |
| (e) M is the CIO, and N is the CEO. |

- 10) Organizations are **most** likely to achieve the expected return on investment (ROI) from information technology investments when the IT investment is accompanied by:

- (a) the latest hardware and software upgrades and high-quality after-sales support.
- (b) increased spending on IT security, security audits, and quality-standards compliance.
- (c) strong complementary assets such as skilled personnel and redesigned business processes.
- (d) reduced risk by outsourcing all IT operations to external organizations and individual suppliers.
- (e) strict reduction of indirect costs such as electricity and other non-IT operational and maintenance costs.

- 11) Electronic commerce (e-commerce) primarily refers to:

- (a) electronic transfer of funds between different types of financial institutions.
- (b) buying and selling goods and services over electronic networks, especially the internet.
- (c) commercial transactions carried out using hand-held mobile devices and wireless networks.
- (d) online communication between employees within an organization and financial organizations.
- (e) the use of computers, networks, and information systems to automate internal business processes.

- 12) A company could significantly reduce its dependence on physical stores by operating online 24/7, serving customers globally, and allowing purchases via laptops or smartphones.

Which key advantage of electronic and mobile commerce is **most** evident in the above scenario?

- (a) Elimination of competition
- (b) Reduced need for skilled employees
- (c) Guaranteed increase in product prices
- (d) Complete elimination of operational costs
- (e) Improved customer convenience and global reach

- 13) An organization is considering implementing the following techniques in its e-commerce system:

- A. Dynamic pricing strategies that adjust prices based on demand and competition.
- B. Offline manual accounting and record-keeping of all transactions to ensure security.
- C. Personalized product recommendations based on customer browsing and purchase history.

Which of the above techniques directly contribute(s) to increasing sales and generating profits?

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|-------------------|---------------------|-------------------|
| (a) Only A. | (b) Only A and B. | (c) Only A and C. |
| (d) Only B and C. | (e) All A, B and C. | |

- 14) A manufacturer of office furniture supplies bulk chairs and desks to corporate offices through an online procurement portal. Businesses place large orders and schedule deliveries electronically.

Which of the following types of e-commerce transactions is described above?

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|----------------------------------|----------------------------------|
| (a) B2B – Business to Business | (b) C2C – Consumer to Consumer |
| (c) B2C – Business to Consumer | (d) B2G – Business to Government |
| (e) G2B – Government to Business | |

- 15) Which of the following is **NOT** a critical technology infrastructure component required to support e-commerce and m-commerce?

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|----------------------------|-------------------------------------|
| (a) Database | (b) Reliable network connectivity |
| (c) Secure web server | (d) Standalone desktop applications |
| (e) Payment gateway system | |

- 16) The main purpose of having a Transaction Processing System (TPS) is to:

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| (a) Process routine business activities. |
| (b) Support long-term strategic planning. |
| (c) Focus on external market intelligence. |
| (d) Support top executives in decision making. |
| (e) Provide predictive analytics and simulations. |

- 17) Which of the following is a key characteristic of a Decision Support System (DSS)?

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| (a) Managing payroll and billing operations in an organization. |
| (b) Making automated decisions to overcome security problems. |
| (c) Supporting semi-structured and unstructured decision making. |
| (d) Processing high volumes of day-to-day transactions repeatedly. |
| (e) Generating fixed, periodic reports to share important information. |

- 18) Which of the following statements is **correct** regarding functional information systems in an organisation?

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| (a) Marketing and Manufacturing Systems operate independently and do not exchange operational data. |
| (b) Human Resource Management Systems exchange information mainly with external regulatory bodies. |
| (c) Finance and Accounting Systems receive data from Marketing and Manufacturing Systems. |
| (d) Manufacturing Systems depend solely on Finance and Accounting Systems for production planning. |
| (e) Marketing Systems rely mainly on Human Resource data to forecast customer demand and price changes. |

19) By implementing a cross-functional information system, an organization primarily aims to support:

- (a) a single department by processing specialized operational data.
- (b) end-to-end business activities across multiple functional areas.
- (c) communication with external stakeholders such as customers and suppliers.
- (d) protecting employees' sensitive data and using it mainly for strategic planning.
- (e) increasing efficiency by operating independently of functional information systems.

20) A consulting firm with multiple locations handles similar client projects regularly. Its employees need quick access to past project reports, best-practice documents, and expert insights to solve problems and improve decision-making.

Which type of information system **best** supports the firm's needs described above?

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|---|-----------------------------------|
| (a) Decision Support System | (b) Transaction Processing System |
| (c) Knowledge Management System | (d) Management Information System |
| (e) Enterprise Resource Planning System | |

21) A manufacturer is struggling to ensure that suppliers deliver materials on time, production lines run smoothly, and products reach customers efficiently.

Which of the following systems is **best** suited to managing these processes?

- (a) Executive Support System
- (b) Knowledge Management System
- (c) Supply Chain Management System
- (d) Enterprise Resource Planning System
- (e) Customer Relationship Management System

22) Which of the following **best** describes the capabilities provided by a Business Intelligence (BI) system?

- (a) Automates routine customer communication and service requests.
- (b) Mainly supports operational transaction processing for daily sales.
- (c) Manages supplier relationships and inventory flow across the organization.
- (d) Enables interactive reporting and analytical modeling to support decision-making.
- (e) Focuses exclusively on capturing and sharing organizational knowledge and experience.

23) To show how leads convert to customers and where drop-offs occur in the sales process, a BI (Business Intelligence) analyst should use a:

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| (a) Bubble Chart | (b) Funnel Chart | (c) Heat Map |
| (d) Line Chart | (e) Pie Chart | |

24) Which of the following **best** defines assistive technology systems?

- (a) Internet platforms that provide online educational courses for those applying for staff assistant positions.
- (b) Systems that help individuals with disabilities perform tasks they would otherwise find difficult or impossible.
- (c) Home appliances that assist housekeepers by automating daily chores, such as cleaning, with little or no user input.
- (d) High-speed computers that are used exclusively in research laboratories to process survey or experimental data.
- (e) Systems that are designed to increase the productivity of newly appointed office workers with limited prior experience.

25) Informatics is the combination of information technology with traditional disciplines, such as medicine or science, while considering the impact on:

- (a) religions, culture, and art.
- (b) relationships, values, and ethics.
- (c) accuracy, validity, and reliability.
- (d) people, knowledge, and education.
- (e) individuals, organizations, and society.

26) Which of the following is **NOT** a common facility provided by an expert system in a business firm?

- (a) An inference engine that applies if-then rules to reach conclusions.
- (b) An explanation facility that explains how a solution or decision was reached.
- (c) A knowledge base that stores facts and rules relevant to a specific problem domain.
- (d) A knowledge acquisition facility that allows experts to input and update knowledge.
- (e) An emotional intelligence module that automatically understands human intentions.

27) Which of the following activities is typically performed first in the information systems acquisition process?

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| (a) Selecting a vendor | (b) Conducting system testing |
| (c) Training end users | (d) Identifying user requirements |
| (e) Installing hardware and software | |

28) Valuing information system acquisitions is initially justified through:

- (a) unit testing of system components.
- (b) cost-benefit analysis based on expectations.
- (c) customer feedback collected after deployment.
- (d) user satisfaction assessment during system use.
- (e) performance evaluation after system implementation.

29) Information systems enable organizational processes **most** directly by:

- (a) automating, integrating, and coordinating business activities.
- (b) focusing exclusively on data storage and information sharing.
- (c) making decisions automatically by replacing all human thinking.
- (d) increasing hardware performance by reducing energy consumption.
- (e) eliminating organizational structures through enhanced communication.

30) The *productivity paradox* of information systems describes that:

- (a) replacing labour with automated processes is difficult.
- (b) employees cannot be easily trained to use new systems.
- (c) implementing and monitoring IT governance is challenging.
- (d) maintaining legacy systems is required to sustain productivity.
- (e) IT investments often do not show immediate productivity gains.

31) Which of the following techniques is **most** suitable for evaluating information system investments when intangible benefits and innovation are important?

- (a) Payback period
- (b) Net present value
- (c) Cost-benefit analysis
- (d) Return on investment
- (e) Multi-criteria analysis

32) Consider the following two lists of information system “sourcing methods” and “descriptions” that have not been correctly matched.

Sourcing Method	Description
A. In-sourcing	I. Relocating IS operations to another country, often to reduce costs.
B. Outsourcing	II. Using internal staff and resources within the organization to develop an IS.
C. Co-sourcing	III. Hiring an external service provider to manage part or all of the IS functions.
D. Offshoring	IV. Partnering with an external vendor to share responsibility for IS functions.

Which of the following correctly matches the above listed “sourcing methods” with the “descriptions”?

- (a) A–II, B–III, C–IV, D–I
- (b) A–III, B–II, C–I, D–IV
- (c) A–IV, B–III, C–I, D–II
- (d) A–I, B–III, C–II, D–IV
- (e) A–II, B–IV, C–III, D–I

- 33) ABC Manufacturers is a construction material distribution company that intends to acquire an information system (IS) to support its ongoing business activities. However, the company currently has no IT experts, and its executives are not fully aware of the information system functionalities they need.

As a consultant, which of the following options do you recommend as the **most** appropriate IS acquisition method for this company?

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|--|--|
| (a) Conduct end-user development | (b) Subscribe to a cloud-based system |
| (c) Borrow software from a competitor | (d) Get the system developed by a vendor |
| (e) Buy proprietary software with one-time payment | |

- 34) Which of the following practices is the **most** ethical behavior in an organization, assuming there is **no** explicit employee consent for monitoring or accessing personal work data?

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| (a) Management reviewing files stored in employees' shared online workspaces. |
| (b) Requiring employees to use activity-monitoring software during working hours. |
| (c) Management periodically reviewing employees' office emails to ensure appropriate use. |
| (d) Allowing employees to take printed documents and electronic files outside the organization. |
| (e) Allowing employees to use company-owned laptops at home in accordance with company policy. |

- 35) Which of the following is **NOT** a principle of technology ethics as defined in the Code of Professional Conduct by the Association of Information Technology Professionals (AITP)?

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|----------------------|-----------------------|---------------------|
| (a) Justice | (b) Proportionality | (c) Minimizing risk |
| (d) Informed consent | (e) Work on schedules | |

- 36) AI systems within an organization are generally restricted from accessing:

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| (a) company-owned data stored in the organization's data warehouse. |
| (b) personal or non-work-related data stored on employees' private computers. |
| (c) user data voluntarily shared through the company's official online platforms. |
| (d) aggregated network usage data generated by employees during official work activities. |
| (e) customer data collected with their consent through online forums and data-collection forms. |

- 37) In computer security, a *buffer overflow* occurs when a:

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| (a) program encrypts data without a key. |
| (b) network buffer exceeds its maximum bandwidth. |
| (c) program runs out of hard drive space while saving files. |
| (d) user tries to log in with the wrong password multiple times. |
| (e) program writes more data into a memory buffer than it can hold. |

38) To ensure the *customer data privacy*, an organization should:

- (a) delete all raw data when a customer requests it.
- (b) inform customers about selected parts of its data policy.
- (c) avoid saving any secured personal data such as bank card details.
- (d) enforce clear rules for collecting, storing, and handling customer data.
- (e) keep secret that customer data is being collected whenever they access the system.

39) Which of the following techniques is **NOT** commonly used in robotics and AI?

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|----------------------|----------------------------|---------------------------------|
| (a) Data mining | (b) Data analytics | (c) Natural Language Processing |
| (d) Machine learning | (e) Database normalization | |

40) Which of the following is **NOT** a common health or safety issue associated with prolonged use of information systems?

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|----------------------------|------------------------------|----------------------|
| (a) Dry eyes | (b) Dementia | (c) Sleep disruption |
| (d) Carpal tunnel syndrome | (e) Repetitive strain injury | |
